



Figure 1: Oracle Berkeley DB

devices, and those capable of communicating with a central database server or other mobile clients from remote sites and handling local queries without connectivity.

Mobile DBMSs should satisfy the following requirements:

- Small memory footprint
- Flash-optimised storage system
- Data synchronisation
- Security
- Low power consumption
- Self-management
- Strong coordination via embeddable features in applications

There are many mobile databases available in the market with many new ones also emerging, but it is the developer who has to choose the right database that satisfies all the requirements of the particular mobile app being created.

A few popular open source databases designed especially for mobile apps are discussed here.

SQLite

SQLite database is a small, highly reliable, embedded, fully-featured, compact and self-contained relational DBMS available as a public domain software package. As compared to other database management systems, SQLite is not a client-server database engine. It is regarded as a popular database, especially for application software like Web browsers and even for mobile app developers to store data from front-end mobile apps. SQLite offers an amazing set of tools to handle all sorts of data with ease and fewer constraints, compared to hosted and process based server relational databases.

Features

- **Serverless:** It doesn't require a separate server process or system to operate. Its library has direct access to storage files.
- **Zero configuration:** No server is required, which means 'No setup'. Creating a SQLite database instance is as easy as opening a simple file.
- **Cross-platform:** It is available for various platforms like Android, BSD, iOS, MAC, Solaris, Windows and VxWorks.

- **Self-contained:** The entire database is contained in a single library and is integrated into the host application.
- **Transactional:** SQLite transactions are fully ACID-compliant, which allows safe access from multiple processes.
- **Reliability:** SQLite development is under consistent development and proper testing is done before launching new versions.
- **Fully-featured:** It contains all the query language features found in the SQL2 standard.
- **SQL extensions:** SQLite provides a number of enhancements to SQL, which are not normally found in other database engines. These enhancements include the EXPLAIN keyword and manifest typing. It also provides statements like REPLACE and the ON CONFLICT clause that allows for adding control over the resolution of the constraint conflict. It also supports the ATTACH and DETACH commands.

Official website: <https://www.sqlite.org/>

Latest version: 3.22.0

ORMLite

Object Relational Mapping Lite (ORMLite) facilitates simple, lightweight Object Relational Mapping (ORM) functionality for persisting Java objects to SQL databases by avoiding the overhead of other ORMs. It simplifies complicated SQL operations via a flexible query builder and provides powerful abstract Database Access Object (DAO) classes.

ORMLite supports JDBC connections to MySQL, Postgres, H2, SQLite, Derby, HSQLDB and Microsoft SQL Server, and can also be extended to various other databases. It supports native database calls on Android OS. ORMLite is bulkier than SQLite or Realm, but slower than both of them. It is regarded as faster than other ORM based databases.

As there is no JDBC support in Android OS, ORMLite makes direct calls to Android database APIs to access SQLite databases.

Features

- Provides a set of lightweight C# extension methods around .NET System.Data.* interfaces.
- High performance database with support for indexes, text blobs, etc.
- Supports cross-platform multiple databases like SQL Server, SQLite, MySQL, PostgreSQL and Firebase running on both .NET and Mono platforms.
- Auto generates SQL to create and drop database tables.
- Has provisional support for DB2, Oracle, ODBC and Netezza.
- Has powerful abstract Database Access Object (DAO) classes.

Official website: <http://ormlite.com/>

Latest version: 5.0